A Byte-Level Neural Morphological Analyser for Classical Ge'ez

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Abstract. We fine-tune ByT5-small to analyse Classical Ge'ez words: given a word, the model writes its consonantal root, its grammatical features and its part of speech. Training uses 137,446 word–analysis pairs from a curated Ge'ez morphology dataset (hand-checked paradigm tables, forms transcribed from printed grammars and dictionaries, and rule-generated forms). Validation and test words are split by root, so no test root is seen in training. On 2,310 held-out test words the analyser identifies the root of 89.8%, produces the complete analysis for 67.4%, and assigns the correct part of speech to 85.7%. Errors concentrate on infinitives, on the noun/adjective distinction and on closed-class words that are rare in training.

1. Task

Ge'ez builds words by interleaving a consonantal root with vowel patterns and affixes. The root s-g-d ('bow, worship') gives ሰገደ 'he bowed', ሰገዱ 'they bowed' and ትሰግዱ '(that) you (pl.) bow'. The analyser maps a word to a text analysis, for example ሰገዱ → *root:* ሰ-ገ-ደ | *tense: past* | *person: 3* | *gender: m* | *number: p* | *stem: basic* (the model's actual output). It is trained on three tasks marked by a prefix: **analyse** (word → root and features), **generate** (root and features → word) and **part of speech** (word → word class). Generation is used only as an auxiliary training task; we report analysis and part of speech.

2. Data

All examples come from the Ge'ez morphology dataset (geez-dataset, snapshot 04f45a7 of 16 October 2026). Every root belongs to exactly one of train, validation and test; validation and test therefore measure analysis of words whose root was never seen.

Source	Word pairs	Description
synthetic	118,131	forms generated from checked roots by paradigm rules
unified book	10,002	forms transcribed from printed grammars and books
gold nominal	6,168	hand-checked noun and adjective paradigm cells
gold verb	1,727	hand-checked verb paradigm cells
dictionary	1,080	dictionary headwords with checked roots
extra synthetic	247	additional checked generated forms
dictionary verb	59	dictionary verbs with checked roots
word without root	32	closed-class words analysed as rootless
total	137,446	562,403 training examples over the three tasks

The held-out sets contain 1,736 validation words (147 roots) and 2,310 test words (217 roots); training covers 1,946 roots. A leakage check confirms that no validation or test root, word form, or spelling variant of a root occurs in training. Verbs make up 75% of training examples and generated forms 86% of word pairs.

3. Model and training

Setting	Value
Model	ByT5-small (byte-level T5): 12 encoder and 4 decoder layers, hidden size 1472, feed-forward 3584, 6 heads; initialised from the public checkpoint
Optimiser	Adafactor, learning rate 1e-3, 500 warm-up steps, linear decay to 1e-4
Batch and length	512 examples per step; 10,980 steps = 10 epochs
Precision and hardware	fp32, data-parallel on two NVIDIA T4 GPUs, about 33 hours
Model selection	validation every 1,000 steps; score = mean of exact-analysis and root accuracy
Decoding	greedy, up to 129 output bytes

Training loss fell from 5.67 to 0.009. Validation accuracy rose quickly and then levelled off (Figure 1); the best validation score (0.809) was reached at step 5,000, and that checkpoint is used for all results below. The final checkpoint scores 0.803, so the second half of training did not improve held-out words.

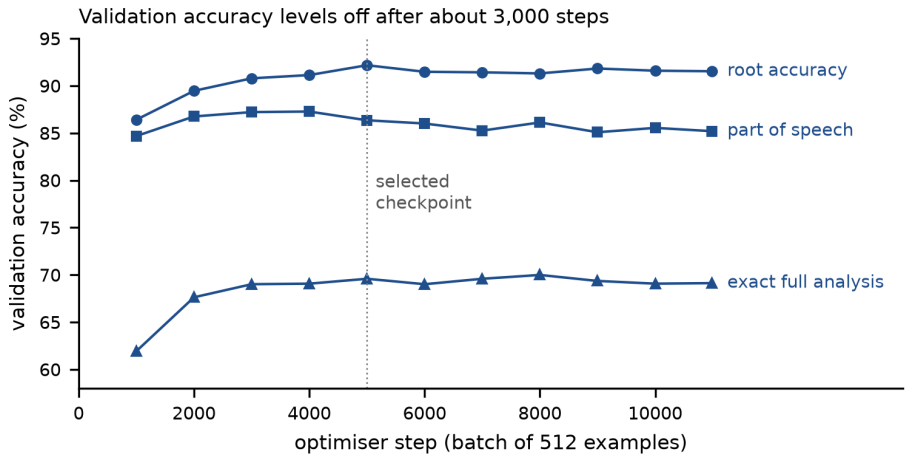


Figure 1. Validation accuracy (1,736 held-out words) every 1,000 steps. The dotted line marks the selected checkpoint (step 5,000).

4. Results

Table 1. Accuracy of the selected checkpoint. *Root*: predicted root equals the gold root. *Exact*: root and every labelled feature are correct. Test rows are split into gold paradigm cells and forms from other printed books.

Set	Words	Root	Exact analysis	Part of speech
validation	1,736	92.1%	69.6%	86.3%
test: gold paradigms	1,543	89.7%	73.7%	87.1%
test: other books	767	90.1%	54.8%	82.9%
test: all	**2,310**	**89.8%**	**67.4%**	**85.7%**

Table 2. Test accuracy by word class (classes with at least 5 test words).

Class	Words	Root	Exact analysis	Part of speech
gold paradigms				
adjective	627	90.4%	73.0%	88.4%
noun	580	86.7%	72.1%	79.8%
verb	336	93.5%	77.7%	97.3%
other books				
verb	363	91.2%	73.6%	96.4%
adjective	259	90.3%	39.4%	87.6%
noun	82	87.8%	37.8%	48.8%
infinitive	28	92.9%	17.9%	0.0%
particle	9	100.0%	66.7%	77.8%
pronoun	8	87.5%	50.0%	87.5%
preposition	6	100.0%	50.0%	50.0%
nominal	5	100.0%	40.0%	40.0%

Test accuracy is close to validation accuracy (2.3 points lower on roots and 2.2 on exact analysis); part of the gap is expected because validation selected the checkpoint. Roots are recovered equally well for both sources (89.7% and 90.1%), but complete analyses are much harder for book forms (54.8%) than for paradigm cells (73.7%): book forms are more varied and carry feature combinations that the paradigm tables do not cover.

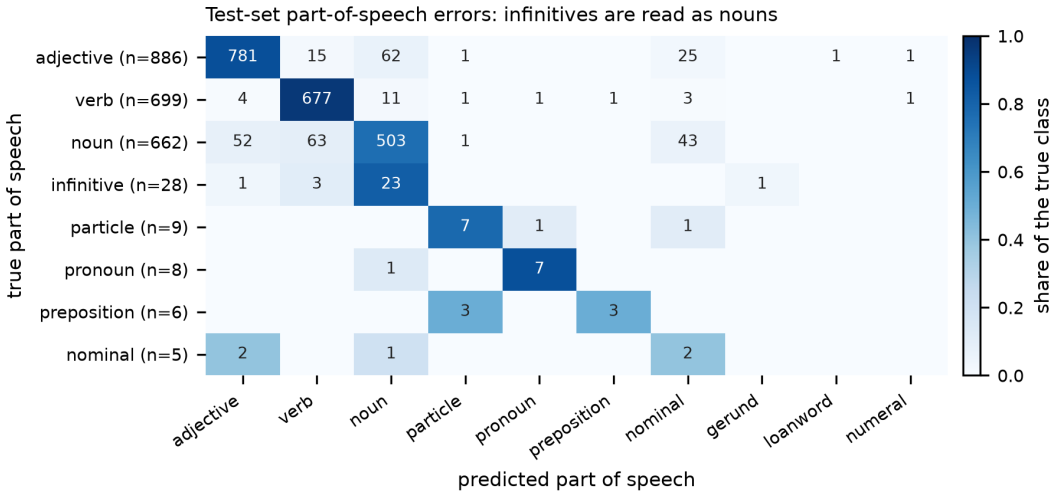


Figure 2. Part-of-speech confusion on the test set (classes with at least 5 words; counts in cells, colour = share of the true class).

5. Error analysis

Three error types dominate. (1) **Infinitives** are tagged as nouns in 23 of 28 test cases, and their full analysis is rarely right; they are rare in training (0.3% of examples) and share their surface shape with nouns. (2) **Nouns** are the hardest content class: 52 of 662 test nouns are tagged as adjectives and 63 as verbs, and 62 of 886 adjectives are tagged as nouns; the noun/adjective distinction often depends on context rather than form. (3) **Closed-class words** (pronouns, prepositions, particles, demonstratives) each make up less than 0.2% of training examples and have too few test words for reliable estimates; prepositions, for example, are split between preposition and particle (3 of 6 correct). Root identification is the most robust part of the output, which makes the model suitable as a source of root labels for downstream tasks.

6. Limitations

Evaluation words are dictionary and paradigm forms out of context; accuracy on running text, where clitics, names and loanwords are common, is likely lower and has not been measured. Labels follow the conventions of the printed sources, and the gold annotation itself may contain errors. Each word is analysed independently, so homographs receive a single analysis. Results come from a single training run.

7. Availability

Code: geez-morphology-model (snapshot a1ad2b6). Data: geez-dataset (snapshot 04f45a7). The trained model, the evaluation script and per-word test predictions accompany this report.